Group Project

[LLSMS2225] Derivatives Pricing

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Outline

1 Project Description

2 Instructions

3 Details

Introduction

- Suppose you are a trader working on a bank's option trading desk.
- Your role is to
  1. Interact with the customer and design option deals.
  2. Implement the deals by trading the options and/or the underlying.
  3. Hedge the deals.
- A customer comes to you with the belief that the price of an asset, we denote it S, will exhibit **heavy variations** in the upcoming year. And she wants you to design a deal to profit off of that.
- Your objective is to (i) design this deal, (ii) price it, (iii) understand the price drivers, and (iv) design a dynamic replicating strategy.

Pricing 101: Calls & Puts

- European Call Option
  - gives the right to the option buyer to **buy** 1 share of S, the **underlying**, at a specific time $T > 0$ and at a specific strike price K. Its payoff is

$$\psi_T^{call} = \psi^{call}(S_T) \quad \text{where} \quad \psi^{call}(x) = \max(0, x - K) \tag{1}$$

- European Put Option
  - gives the right to the option buyer to **sell** 1 share of S, the **underlying**, at a specific time $T > 0$ and at a specific strike price K. Its payoff is

$$\psi_T^{put} = \psi^{put}(S_T) \quad \text{where} \quad \psi^{put}(x) = \max(0, K - x) \tag{2}$$

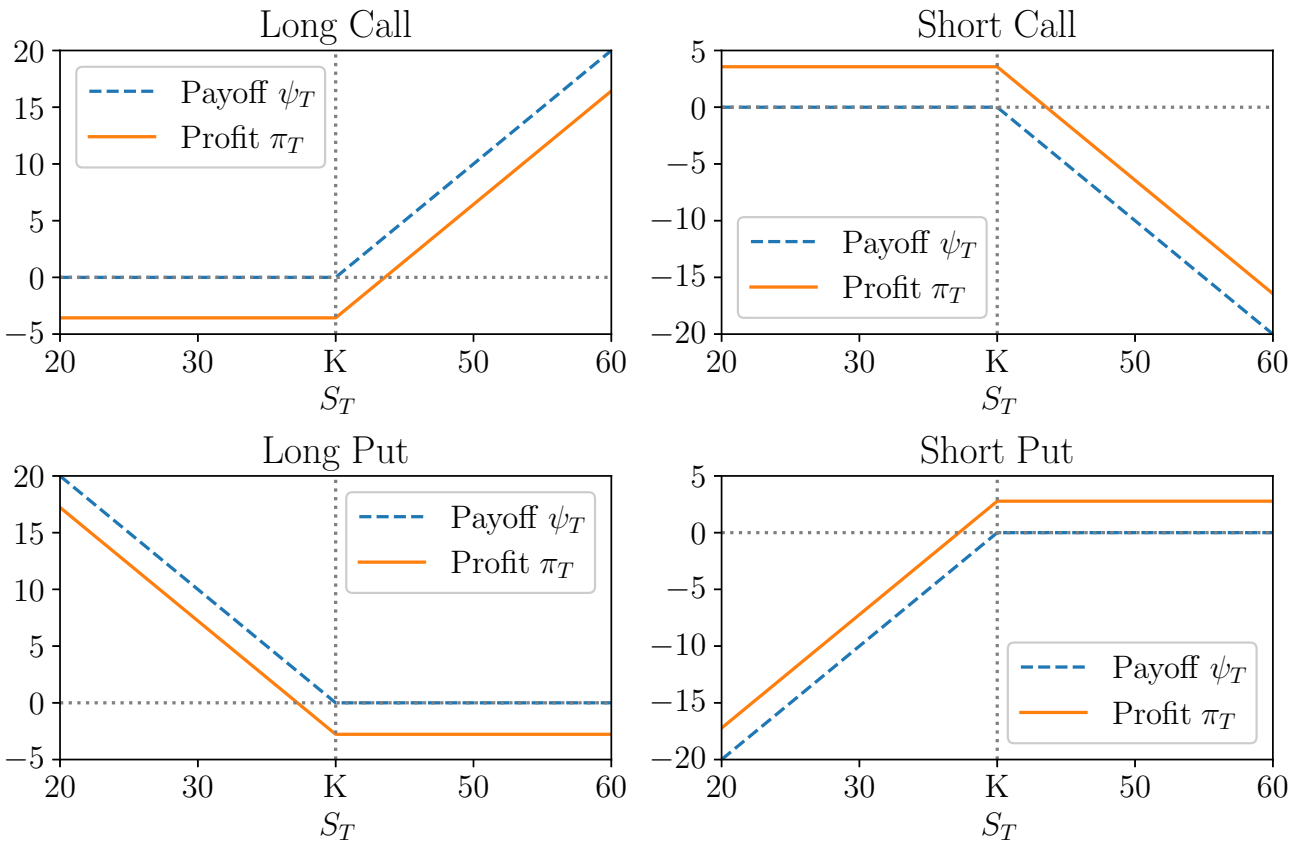
- The **profit** π_T for a purchased option is given by

$$\pi_T^{option} = \psi_T^{option} - V_0^{option}, \tag{3}$$

where V_0^{option} is the option price at time $t = 0 < T$

Call and Put payoff and profit functions

Figure: Call and Put payoff and profit functions



Notes. Profit and Payoff functions for european calls and puts. We set $K = S_0 = 40$, $\sigma = 0.2$, and $T = 1$.

Q1 : Construction and Intuition of the trade

- You understand that you must design a deal that delivers a positive payoff ψ_T increasing as S_T moves away from the strike price $K = S_0$ in either direction, i.e., when volatility is high.
- Two possible payoff structures could be *absolute* and *quadratic*:

$$\psi_T^a = |S_T - K| \quad \text{and} \quad \psi_T^q = (S_T - K)^2 \tag{4}$$

- The customer is not sure which is best. Your role is to advise her.

Q1 : Construction and Intuition of the trade

- *A priori,* two payoff structures could be used, *absolute* and *quadratic*:

$$\psi_T^a = |S_T - K| \quad \text{and} \quad \psi_T^q = (S_T - K)^2 \tag{5}$$

- Q1.1. How do you construct these deals from vanilla options? Can you use standard calls and puts for both of them?
- Q1.2. Plot the payoff functions and compare them. What are the main differences?
- Q1.3. Intuitively, without performing any calculations, how would you expect the option prices V_0^a, V_0^q to compare? Under what conditions do you anticipate $V_0^a < V_0^q$ and conversely?

Options 101: Pricing

- The time-t price of a European option can be written as

$$V_t = \mathbb{E}^{\star}[\psi_T | \mathcal{F}(t)] e^{-r(T-t)} \quad \forall t \leq T \tag{6}$$

- Three alternatives for pricing:
  1. Monte-Carlo simulations
     - ★ Pros: Easy to implement and code
     - ★ Cons: Computationally expensive
  2. Numerical integration
     - ★ Pros: No need for analytical formula
     - ★ Cons: Computationally expensive/sensitivity in high dimension
  3. Analytical pricing formulas (see Black-Scholes-Merton (BSM) model)
     - ★ Pros: Computationally inexpensive
     - ★ Cons: Difficult to derive a formula, sometimes impossible

Q2 : Pricing

Consider two options, with payoffs ψ_T^a and ψ_T^q. For both options:

- Q2.1. Compute the option price using Monte Carlo (MC) simulations.
- Q2.2. Compute the option price using numerical integration.
- Q2.3. Can the price of this option be derived analytically? If so, describe the analytical approach and compare the analytical prices with those obtained via MC simulations and numerical integration.
- Q2.4. Analyze (and plot) how the option price varies with key parameters such as the initial underlying price S_0, time to maturity T, strike price K, and volatility of the underlying asset σ.
- Q2.5. Can the option's Delta (Δ_t) be computed analytically? If yes, discuss its possible values and behavior.

Q2 : Pricing (ctd.)

- Q2.6. Plot on 4 graphs (i) the payoff function ψ_T, (ii) the price function V_t at three different times $t \in (0, T/2, T)$, (iii) the profit function $\psi_T - V_0$ and (iv) the delta function Δ_t at three different times $t \in (0, T/2, T)$, as a function of S_T.

Compare the results between both options as well as with a standard european call. What do you observe?

Q3 : Replication

- Q3.1. For both payoff structures ψ_T^a and ψ_T^q, describe and implement a dynamic replication strategy replicating the payoff at maturity.
- Q3.2. In real life, every transaction (buying/selling stocks) made on an exchange comes at a cost, known as the transaction cost. Assuming a transaction cost of $c = 50bps$ per dollar traded, what is the impact on your replicating strategy?
- Q3.3. Compare the performance and behavior of the two replication strategies. Based on this comparison, and considering the key differences between the two option contracts, what option would you sell to your client?

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Support

1. Office hours: **6 hours** per week on Tuesdays and Fridays, 20mins/group, booking with Moodle. Take advantage of this opportunity!
2. I am available via email at rodolphe.vanderveken@uclouvain.be.

Instructions

1. Groups: 4 per group (max). Groups must be determined by 11/12 11.59pm.
2. If you do not find a group, **send both of us an email** before the deadline. Random groups will be formed for the remaining students / we will assign remaining students to existing groups.
3. **You must be in a group**. If you are not, you will not obtain a grade for the project.

Instructions: Evaluation

1. Project report: 30% of the final grade of this course
2. Individual oral exam based on the project: 10% of this course. Questions on implementation (code) and content.
3. Upload pdf report + R/Python code (**clear**, **commented** and **ready to run**, see Details section).
4. Deadline: **11/01/2026 11.59pm**
5. Note that to report the note of the project, you must present the individual oral exam.
6. If you fail the project report in June, you have the possibility to send an improved version in August.

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Project Goals

- *Autonomy*. The content of the course alone might not be sufficient to complete the project. Anticipate the course schedule, and start early!
- *Ambition*. Do not hesitate to go beyond what is asked of you and provide relevant additional analysis. Discuss and explain the results that you obtain, do not just report them.
- *Understanding*. Show that you understand the theoretical concepts, the mathematics behind, and the results that you obtain.
- *Clarity*. Quality of mathematical specifications, conciseness, display of results.

Instructions: code

1. **In one click**, we must be able to see
   - All figures in your report.
   - Plot(s) for one illustrative path with: option price, value of replicating portfolio, rebalancing dates, evolution of Δ_t, value of cash account and equity account.
2. Clear and commented parameter setup (strike price, volatility, time to maturity, rebalancing frequency, etc.) which can be played with.
3. On top of that, we must be able to reproduce all analysis mentioned in the .pdf report.

Assumptions

1. Suppose that you buy and sell at the market fair price.
2. Suppose that your observations of the price are equally spaced for some discretization step $\delta > 0$:

$$t_i = i\delta, \quad i \in 1, 2, ..., n, \quad n = T/\delta \tag{7}$$

3. Assume S follows a GBM (Geometric Brownian motion):

$$dS_t = rS_t dt + \sigma S_t dWt, \tag{8}$$

   where r is the risk-free rate, σ is the volatility of the underlying, and $W = (W_t)_{t \geq 0}$ is a Brownian motion under the R-N measure $\mathbb{P}^\star$
4. Under these assumptions, trajectories of the S process at times $t_1, t_2, ..., T$ can be simulated using the Euler scheme (see Ch14 of reference book)

Assumptions (ctd.)

1. Suppose $S_0 = 30$ and $K = 30$, i.e., the payoff is defined by deviations from the starting price.
2. Suppose the annualized volatility of S is $\sigma = 0.15$.
3. Consider TTM=1y, but feel free to discuss the impact of TTM on your results.
4. Consider an annualized risk-free rate of $r = 4\%$ (10Y U.S. Treasury Bonds).

Options 101: Replication and hedging

For Q3, it is **highly recommended** to read the reference book.

- A **replication strategy** replicates the payoff of the option at maturity without purchasing any option - only by trading the underlying.
- The "Delta" (Δ) of an option is the 1st derivative of the option price relative to the price of the underlying. It changes over time.
- Δ can be used to define a "Δ-hedging strategy"
- In theory, if you rebalance continuously in $[0, T]$, it is possible to replicate exactly the payoff.
- In practice, as you work in discrete time, you will make hedging errors.